

1. Taylor Series for $\sin, \cos$.

$$\sin x = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} x^{2m+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \quad (\forall) x \in \mathbb{R}.$$

$$\cos x = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} x^{2m} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots, \quad (\forall) x \in \mathbb{R}.$$

$$\sin x = \sum \quad (\text{see why in lecture notes})$$

$$\begin{aligned} \cos x = f(x) &= \sum_{m=0}^{\infty} \frac{f^{(m)}(x_0)}{m!} (x-x_0)^m \stackrel{x_0=0}{=} \sum_{m=0}^{\infty} \frac{f^{(m)}(0)}{m!} x^m \stackrel{m=2k}{=} \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \quad \text{without } (2k+1) \text{ because that is } 0. \end{aligned}$$

$$f(0) = \cos 0 = 1$$

$$f'(0) = -\sin 0 = 0$$

$$f''(0) = -\cos 0 = -1$$

$$f^{(3)}(0) = \sin 0 = 0$$

$$f^{(4)}(0) = \cos 0 = 1$$

↓

$$\text{Pattern: } f^{(2k+1)}(0) = 0$$

$$f^{(2k)}(0) = (-1)^k$$

$$\Rightarrow \cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots, \quad (\forall) x \in \mathbb{R}.$$

- For what kind of x does this converge? -

Recall: $\sum_{n=0}^{\infty} a_n (x-x_0)^n$ power series

$$L = \lim_n \left| \frac{a_{n+1}}{a_n} \right| = \lim_n \sqrt[n]{|a_n|}$$

$$R = \frac{1}{L} \text{ radius of convergence.}$$

$$\Rightarrow \sum \dots \text{ abs. conv. } (\forall) x \in (x_0 - R, x_0 + R)$$

$$\Rightarrow \sum \dots \text{ div.}$$

$$a_{2k} = \frac{(-1)^k}{(2k)!}$$

(you could also take $t = 2k \dots$)

$$L = \lim_{k \rightarrow \infty} \left| \frac{a_{2(k+1)}}{a_{2k}} \right| = \lim_{k \rightarrow \infty} \left| \frac{(-1)^{2k+2}}{(2k+2)!} \cdot \frac{(2k)!}{(-1)^k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(-1)^{k+2}}{(2k+1)(2k+2)} \right| =$$

$$= \lim_{k \rightarrow \infty} \frac{1}{(2k+1)(2k+2)} = 0$$

$$\Rightarrow R = \frac{1}{L} = \frac{1}{0} = \infty$$

$$\Rightarrow (0-R, 0+R) = (-\infty, \infty) = \mathbb{R} \Rightarrow \text{abs. conv.}, (\forall) x \in \mathbb{R}.$$

Abs. conv. series $\Rightarrow$ we can $\int$, diff.

($\sum x_n$ abs. conv. if $\sum |x_n|$ conv.)

$\cos x$ is abs. conv., $(\forall) x \in \mathbb{R}$ diff.

$$\begin{aligned} (\cos x)' &= \left(\sum_{k=0}^{\infty} \frac{(-1)^k \cdot x^{2k}}{(2k)!} \right)' \xrightarrow{\text{abs. conv.}} \sum_{k=0}^{\infty} \left(\frac{(-1)^k \cdot x^{2k}}{(2k)!} \right)' \xrightarrow{\text{diff. term by term}} \\ &= \sum_{k=0}^{\infty} (-1)^k \cdot \left(\frac{x^{2k}}{(2k)!} \right)' = \sum_{k=1}^{\infty} (-1)^k \cdot \frac{2k \cdot x^{2k-1}}{(2k)!} = \sum_{k=1}^{\infty} (-1)^k \cdot \frac{x^{2k-1}}{(2k-1)!} \end{aligned}$$

for $k=0$: $(-1)^0 \cdot \frac{x^0}{0!} = 1 \Rightarrow$ no contribution to the sum

$$((2k)!)' = (2k)!$$

$$= -x + \frac{x^3}{3!} - \frac{x^5}{5!} + \dots$$

$$= - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right)$$

$$= -\sin x$$

$$\Rightarrow (\cos x)' = -\sin x.$$

$$\boxed{\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots = \sum_{k=0}^{\infty} (-1)^k \cdot \frac{x^{2k+1}}{(2k+1)!}}$$

$$(\sin x)' = 1 - \frac{3x^2}{3!} + \frac{5x^4}{5!} - \dots = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots = \cos x.$$

a) We proved that $(\sin x)' = \cos x$
 $(\cos x)' = -\sin x$

b) Deduce that: $x - \frac{x^3}{6} < \sin x < x$, $(\forall) x > 0$.
 $1 - \frac{x^2}{2} < \cos x < 1 - \frac{x^2}{2} + \frac{x^4}{24}$

Recall Taylor's formula:

$$f(x) = T_m(x) + R_m(x)$$

$$T_m(x) = \sum_{k=0}^m \frac{f^{(k)}(x_0)}{k!} \cdot x^k, \quad x_0 = 0 \text{ (here)}$$

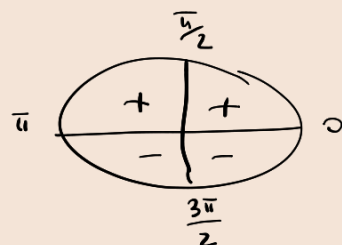
$$R_m(x) = \frac{f^{(m+1)}(c)}{(m+1)!} \cdot x^{m+1}$$

$\Rightarrow$ like the next term in the Taylor's polyn.
 but with c intermediate value. $c \in (0, x)$

$$T_1(x) = x$$

$$T_2(x) = x$$

$$T_3(x) = x - \frac{x^3}{3!}$$



Method 1

$$f(x) = T_3(x) + R_3(x) = x - \frac{x^3}{3!} + \underbrace{\frac{f^{(4)}(c)}{4!} \cdot x^4}_{>0}$$

$$f^{(4)}(c) = \sin c > 0, \quad (\forall) c \in (0, \pi)$$

$$(c \in (0, x))$$

$$\Rightarrow f(x) > x - \frac{x^3}{3!}, \quad (\forall) x \in (0, \pi)$$

?

$$\frac{x > \pi}{x - \frac{x^3}{3} < \pi - \frac{\pi^3}{3!} < -1 < \sin x.}$$

Method 2

$$f(x) = T_2(x) + R_2(x) =$$

$$= x + \frac{f^{(3)}(c)}{3!} \cdot x^3$$

$$= x - \underbrace{\frac{\cos c \cdot x^3}{3!}}_{<}$$

$$-\cos x \leq 1.$$

$$-\cos c \leq 1$$

$$\Rightarrow f(x) \geq x - \frac{x^3}{3!}$$

c) Prove formally Euler's formula $e^{ix} = \cos x + i \cdot \sin x$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad x \mapsto ix$$

$$e^{ix} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} = 1 + \boxed{ix} + \frac{(-1) \cdot x^2}{2!} + \frac{(-i) \cdot x^3}{3!} + \frac{x^4}{4!} + \frac{i \cdot x^5}{5!} + \dots$$

$$= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) + i \cdot \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)$$

$$= \cos x + i \cdot \sin x$$

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = -i$$

$$i^4 = 1$$

! Power series makes sense with $\mathbb{C}$ nr. as well!

② a) For $\alpha \in \mathbb{R}$, $|x| < 1$, prove the binomial series.

$$(1+x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k$$

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1) \dots (\alpha-k+1)}{k!}$$

$$\binom{\alpha}{0} = 1$$

$$\binom{m}{k} = \binom{m}{m-k} = \frac{m!}{k!(m-k)!} = \frac{m(m-1)(m-2) \dots (m-k+1)}{k!}$$

"m choose k"

$$f(x) = (1+x)^\alpha = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = \sum_{k=0}^{\infty} \frac{\alpha(\alpha-1) \dots (\alpha-k+1)}{k!} x^k$$

$$f'(x) = \alpha(1+x)^{\alpha-1}$$

$$f''(x) = \alpha(\alpha-1)(1+x)^{\alpha-2}$$

⋮

$$f^{(m)}(x) = \alpha(\alpha-1) \dots (\alpha-m+1)(1+x)^{\alpha-m} \rightarrow \text{Pattern}$$

$$\hookrightarrow f^{(m)}(0) = \alpha(\alpha-1) \dots (\alpha-m+1)$$

$$f(x) = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k$$

$\downarrow$
 a_k

- I have completed the Taylor series -

- Is it convergent? If yes, it will converge to $f(x)$ -

$$L = \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{\alpha(\alpha-1) \dots (\alpha-k+1)(\alpha-k)}{k!(k+1)} \cdot \frac{k!}{\alpha(\alpha-1) \dots (\alpha-k+1)} \right| =$$

$$= \lim_{k \rightarrow \infty} \left| \frac{\alpha-k}{k+1} \right| = |-1| = 1 > 0$$

$$\Rightarrow R = \frac{1}{L} = \frac{1}{1} = 1. \Rightarrow \text{abs. conv, } (\forall) x \in (-1, 1).$$

b) Find the first 4 terms in the expansion of $\sqrt{1+x}$, $\frac{1}{\sqrt{1+x}}$

$$(1+x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k$$

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1) \dots (\alpha-k+1)}{k!}$$

$$\binom{\alpha}{0} = 1 = C_\alpha^0$$

$$\sqrt{1+x} = (1+x)^{\frac{1}{2}} \quad \underline{\alpha = \frac{1}{2}} \quad \sum_{k=0}^{\infty} \binom{\frac{1}{2}}{k} x^k$$

$$= 1 + \binom{1/2}{1} x + \binom{1/2}{2} x^2 + \binom{1/2}{3} x^3 + \dots$$

$$= 1 + \frac{\frac{1}{2}}{1!} x + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!} x^2 + \frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)}{3!} x^3$$

$$\binom{1/2}{1} = \frac{\frac{1}{2}}{1!} = \frac{1}{2}$$

$$\binom{1/2}{2} = \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!}$$

$$\frac{1}{\sqrt{1+x}} = (1+x)^{-\frac{1}{2}} \quad \underline{\alpha = -\frac{1}{2}} \quad \sum_{k=0}^{\infty} \binom{-\frac{1}{2}}{k} x^k$$

$$= 1 + \binom{-1/2}{1} x + \binom{-1/2}{2} x^2 + \binom{-1/2}{3} x^3 + \dots$$

$$= 1 + \dots$$

③ Find the Taylor series around 0 and its radius of convergence for:

a) $f(x) = a^x, a > 0.$

$$f'(x) = a^x \cdot \ln a$$

$$f''(x) = (a^x \cdot \ln a)' = a^x \cdot (\ln a)^2$$

⋮

$$f^{(m)}(x) = a^x \cdot (\ln a)^m$$

$$\Rightarrow f^{(m)}(0) = a^0 \cdot \ln^m a = (\ln a)^m$$

$$\Rightarrow a^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \cdot x^n = f(x) = \sum_{n=0}^{\infty} \frac{(\ln a)^n}{n!} \cdot x^n$$

- Where do we get convergence? -

$$a_n = \frac{(\ln a)^n}{n!}$$

$$L = \lim_n \left| \frac{a_{n+1}}{a_n} \right| = \lim_n \left| \frac{(\ln a)^{n+1}}{n!(n+1)} \cdot \frac{n!}{(\ln a)^n} \right| = \lim_n \left| \frac{\ln a}{n+1} \right| = 0.$$

$$R = \frac{1}{L} = \frac{1}{0} = \infty \Rightarrow \sum \text{abs. conv.}, (\forall) x \in \mathbb{R}.$$

b) $f(x) = (1+x) \ln(1+x)$

$$f'(x) = (1+x)' \cdot \ln(1+x) + \cancel{(1+x)} \cdot \frac{1}{\cancel{1+x}} = \ln(1+x) + 1.$$

$$f''(x) = \frac{1}{1+x}$$

$$f^{(3)}(x) = ((1+x)^{-1})' = -\frac{1}{(1+x)^2}$$

$$f^{(4)}(x) = \frac{2}{(1+x)^3}$$

⋮

$$f^{(m)}(x) = \frac{(m-2)! \cdot (-1)^m}{(1+x)^{m-1}}, \quad (\forall) \underline{m \geq 2}, m \in \mathbb{N}$$

✓ the pattern applies for these numbers.

$m=0$: $f(0) = 1 \cdot \ln 1 = 0$

$m=1$: $f'(0) = 1$

$m \geq 2$: $f^{(m)}(0) = (-1)^m \cdot (m-2)!, m \geq 2.$

$$f(x) = (1+x) \cdot \ln(1+x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \cdot x^n$$

$$\begin{aligned} \text{for } x \in (-1, 1) &= f(0) + f'(0) \cdot x + \sum_{n=2}^{\infty} \frac{(n-2)! \cdot (-1)^n}{n! \cdot n(n-1)} \cdot x^n \\ &= 0 + 1 \cdot x + \sum_{n=2}^{\infty} \frac{(-1)^n}{n(n-1)} \cdot x^n \end{aligned}$$

$\rightarrow a_n$

$$L = \lim_n \left| \frac{a_{n+1}}{a_n} \right| = \lim_n \left| \frac{(-1)^{n+1}}{(n+1) \cdot n} \cdot \frac{n(n-1)}{(-1)^n} \right| = \lim_n \left| \frac{n-1}{n+1} \right| = 1$$

$$\Rightarrow R = \frac{1}{L} = 1. \Rightarrow \sum \text{abs. conv.}, \text{ for } x \in (-1, 1).$$

c) $\sin^2(x) = f(x)$

Method 1

$$f'(x) = 2 \sin x \cos x = \sin 2x$$

$$f'(0) = 0.$$

$$f''(x) = 2 \cos 2x$$

$$f''(0) = 2$$

$$f^{(3)}(x) = -2^2 \sin 2x$$

$$f^{(3)}(0) = 0.$$

$$f^{(4)}(x) = -2^3 \cos 2x$$

$$f^{(4)}(0) = -8$$

Pattern: $f^{(2k+1)}(0) = 0$

$$f^{(2k)}(0) = (-1)^{k+1} \cdot 2^{k-1}$$

$$f(x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot 2^{k-1}}{(2k)!} \cdot x^{2k} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot 2^k}{2 \cdot (2k)!} \cdot x^{2k}$$

($k=0$ does not change the sum)

$$= \frac{1}{2} \cdot \sum_{k=1}^{\infty} (-1)^{k+1} \cdot \frac{(2x)^{2k}}{(2k)!}$$

$$\frac{2^k \cdot x^k \cdot x^k}{(2k)!}$$

Method 2

$$\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x$$

$$1 - \cos 2x$$

$$\Rightarrow \sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\cos 2x = \sum_{k=0}^{\infty} \frac{(2x)^{2k} \cdot (-1)^k}{(2k)!} \quad \xleftarrow{x=2x} \quad \cos x = \sum_{k=0}^{\infty} \frac{(-1)^k \cdot x^{2k}}{(2k)!}$$

$$\sin^2 x = \frac{1}{2} - \frac{\cos 2x}{2}$$

$$= \frac{1}{2} - \frac{1}{2} \cdot \sum_{k=0}^{\infty} \frac{(2x)^{2k} \cdot (-1)^k}{(2k)!}$$

$$= \frac{1}{2} + \frac{1}{2} \cdot \sum_{k=0}^{\infty} \frac{(-1)^{k+1} \cdot (2x)^{2k}}{(2k)!}$$

